Dossier Space Architecture Status at the Session 9 Milestone (By Leonard)

Purpose of this note

This note records how the principal parts of the Dossier Space architecture are currently being used in the Enron pilot, what each component contributes, and whether it is operational, partial or still proposed.

The milestone matters because Session 9 demonstrated that the analytical method can locate and explain an evidential hanging thread, but also exposed that the independent custody record was not produced. The architecture therefore needs to be understood as several connected but distinct layers rather than as a single AI process.

1. The overall architecture

At its simplest, the current architecture is:

Human operator
→ Claude Code host and AI
→ Primers and governing instructions
→ Dossier Space MCP custody server
→ Read-only evidence corpus
→ ROMER finding and session records
→ Independent review and defensibility score

MCP is the interface through which the AI is intended to reach controlled evidence tools. Dossier Space is the wider evidential environment around that interface: custody, provenance, scope, governance, structured reasoning and assurance.

The architecture produces two complementary records:

1. **Infrastructure trace** — what tools were actually invoked, what evidence was accessed, when it was accessed, what hashes applied and what data was returned.

2. **Interpretive trace** — why the AI searched, what it observed, what evidence it relied on, how it reasoned and what conclusion it reached.

The infrastructure trace should be produced by the custody layer. The interpretive trace is primarily produced through ROMER.

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2. Human operator

Architectural role

The operator:

- defines or approves the task;
- approves the relevant primer;
- sets boundaries, custodians, dates and stopping rules;
- authorises material changes in direction;
- challenges the AI where necessary;
- decides whether the resulting finding may be relied upon.

The operator is not expected to perform every search personally. The operator governs the conditions under which the AI works and remains accountable for significant decisions.

Current implementation

This is operational.

In Session 9, the operator approved the ±7-day corroboration window, required a ROMER template before analysis proceeded, selected the direct Moorer candidate from the initial triage and instructed the AI to pause before considering the remaining 26 weaker candidates.

Architectural significance

The operator prevents the AI from silently redefining the matter. This is the principal human control over scope, proportionality and material judgement.

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3. Claude Code

Architectural role

Claude Code is currently the **AI host and working agent**.

It:

- receives the operator's instructions;
- reads the primers and templates;
- formulates and executes searches;
- examines retrieved evidence;

- creates transcripts, findings and supporting files;
- reports uncertainties and deviations;
- pauses for operator decisions.

Claude Code is therefore both an analytical engine and an agent capable of invoking tools.

Current implementation

Operational for direct filesystem analysis.

Session 9 showed that Claude could search the Enron corpus, change search tools after a timeout, deduplicate messages, inspect MIME structure, search for corroboration and prepare a ROMER finding.

However, the session also showed that Claude Code can operate directly through Bash and filesystem tools. When it does so, the work may bypass the intended independent custody layer.

Architectural limitation

Claude's own transcript or JSONL cannot independently prove what Claude did. It is the subject's account of its own work. It is valuable, but it is not the external custody record.

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4. The Meta-Primer

Architectural role

The Meta-Primer is the constitutional layer.

It should contain the standing principles that no task-specific primer may override, including:

- source-derived evidence takes precedence over model recall;
- filed records are frozen;
- provenance divergence must be surfaced;
- no axis may be traded away to compensate for weakness in another;
- corrections are forward-only and made through superseding records;

- conflicts between procedural, reasoning and agentic requirements must be resolved explicitly.

The Meta-Primer governs the interaction between the architecture's axes rather than merely adding another set of instructions.

Current implementation

Conceptually established and reflected in the pilot's behaviour, particularly:

- document is custody, not memory;
- corrections are forward-only;
- deviations are retained rather than erased;
- incomplete infrastructure is recorded rather than concealed.

Current status

Partially operational.

The principles are influencing the pilot, but the final Meta-Primer and its relationship to the individual axial primers still need to be fixed as formal artefacts.

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5. Stage or task primer — CIPDA

Architectural role

The task primer tells the AI how the particular stage of work is to be conducted.

CIPDA provides:

- **Context** — corpus, custodians, limitations and relevant background;
- **Intent** — what is being established and what is expressly outside scope;
- **Plan** — search and analytical method;
- **Deliver** — required outputs;
- **Assure** — evidential boundaries, uncertainty, checks and review requirements.

It converts a broad instruction into an agreed operating method before substantive analysis begins.

Current implementation

Operational.

The Session 9 Analysis primer specified:

- the admitted corpus;
- Farmer as custodian and Moorer as a referenced party;
- the hanging-thread proposition;
- seed phrases and variants;
- the ±7-day corroboration window;
- the candidate classification;
- the required output;
- the stopping condition.

Correction required

The primer incorrectly stated that the absence of a Moorer custodian mailbox meant that messages “from” Moorer could not exist in the corpus. The finding itself disproved that assumption by locating a received copy in Farmer’s mailbox.

The corrected architectural wording should be:

The corpus contains no Moorer custodian mailbox. Messages sent by Moorer may nevertheless exist as received copies in admitted custodians’ mailboxes, but the corpus cannot be treated as a complete collection of Moorer’s communications.

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6. The ROMER template and finding

Architectural role

ROMER is the interpretive record:

- **Reasoning** — why the item matters and what question is being addressed;
- **Observations** — what is directly seen in the material;
- **Method** — what searches, checks and comparisons were performed;
- **Evidence** — the sources and identifiers relied upon;
- **Report/Result** — the conclusion, confidence, limitations and open questions.

ROMER records how an evidential conclusion was formed. It does not independently prove which system actions occurred.

Current implementation

Operational.

F-001 recorded:

- the email and its three duplicate folder copies;
- the literal reference to the phone call and fax;
- the missing attachment;
- the corroboration searches;
- the absence of relevant counterparty mailboxes;
- the distinction between what was observed and what could not be known;
- the resulting hanging thread.

Architectural significance

ROMER made the conclusion challengeable. A reviewer can examine whether the reasoning followed from the declared evidence rather than simply accepting the AI's answer.

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7. The MCP server

Architectural role

The MCP server is the controlled mediation and custody layer between the AI and the evidence.

It is intended to provide tools such as:

- read_file;
- list_folder;
- search_messages;
- get_headers;
- hash_document;

- verify_hash.

The planned server also applies:

- scope and permission checks;
- read-only enforcement;
- privilege controls;
- controlled evidence retrieval;
- source hashing;
- custody logging;
- provenance reporting;
- linkage between evidence access and the relevant ROMER task.

What the MCP server does not do

The MCP server does not decide what the evidence means.

It should not replace:

- the primer;
- the AI's analysis;
- ROMER;
- the operator;
- the final assurance judgement.

MCP provides the standard tool interface. Dossier Space adds the legal and evidential controls around it.

Current implementation

The MCP server was built and made available to Claude Code.

However, in Session 9 the required custody function was not completed:

- the server did not persist an independent access log;
- the Claude JSONL indicates that custody MCP tools were not used for the substantive work;

- the searches were instead carried out through Claude's direct filesystem and Bash tools.

The Session 9 package initially described the server as responding to queries but not writing a persistent log. The later independent review concluded that the custody layer was not demonstrated and that the run could not support trace-versus-log comparison.

Current status

Built but not yet proven as the exclusive evidence-access route.

This is the most important architectural issue before the next scored session.

The next run must establish that:

1. Claude actually invokes the MCP evidence tools;
2. evidence access cannot silently bypass the server;
3. the server writes a persistent independent record;
4. the record can be reconciled against Claude's declared actions.

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8. The read-only Enron evidence corpus

Architectural role

The corpus is the evidence repository.

It should remain:

- locally controlled;
- read-only;
- identifiable and reproducible;
- separated from generated findings and working documents;
- accessible only through governed tools during a fully mediated run.

Current implementation

Operational as a local read-only dataset.

The pilot has established corpus structure, custodians and known limitations. It has also exposed the difference between:

- absence of a custodian mailbox;
- absence of a message;
- absence of an attachment;
- absence of corroboration within a specified search.

Architectural limitation

Read-only status alone is not enough. If Claude can access the corpus directly through unrestricted Bash or filesystem commands, the corpus is protected from alteration but not fully mediated for audit purposes.

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9. The custody log

Architectural role

The custody log should be an append-only or chained record produced independently of the AI's own account.

It should record:

- session and task identifiers;
- tool invoked;
- search terms or request parameters;
- files or records accessed;
- timestamp;
- hashes or source identifiers;
- data returned;
- refusals, scope failures and errors;
- linkage to the relevant primer, proposition and finding.

This is the architecture's answer to the principle:

The watch is never the watched.

Current implementation

Not achieved in Session 9.

The PowerShell transcript, Dossier Space transcript and Claude Code JSONL are valuable supplementary records, but none is a substitute for the independent custody log.

Current status

Critical incomplete component.

This is why Session 9 was judged:

- analytical test — pass;
- architectural test — incomplete;
- overall — qualified pass.

The independent review recommends a preflight gate: no scored run should begin until the server has written a test access entry and that entry has been verified outside Claude's own working record.

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10. PowerShell and Blue Panel transcript

Architectural role

The PowerShell transcript records the operator-side stream of work:

- commands entered;
- visible output;
- session timing;
- operator interventions;
- some errors and warnings.

Current implementation

Operational.

Architectural significance

It helps reconstruct what appeared on the operator's screen and what instructions were given.

Limitation

It does not prove every evidence access, tool invocation or data transfer. It complements the custody log but cannot replace it.

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11. Claude Code JSONL

Architectural role

The JSONL is a detailed record of the Claude Code interaction:

- prompts;
- assistant responses;
- tool requests;
- tool results;
- internal session events.

Current implementation

Operational and useful.

Architectural significance

It allows detailed reconstruction of the AI-side session and can reveal whether MCP tools were available or invoked.

Limitation

It is still generated within the Claude environment. It is therefore an internal trace, not an independent custody record.

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12. Findings and evidence packages

Architectural role

A finding is the reviewable product of an analysis task.

The evidence package should contain:

- the approved primer;
- the relevant evidence identifiers;
- the ROMER finding;

- the session transcript;
- tool and error records;
- custody-log references;
- operator decisions;
- amendments or superseding findings;
- the resulting scorecard.

Current implementation

Substantially operational.

Session 9 produced:

- the primer;
- the ROMER template;
- F-001;
- the session transcript;
- the Claude Code JSONL;
- the Blue Panel record;
- the smoke-test package;
- the independent review.

The package is analytically strong but incomplete because the custody log is absent.

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13. Defensibility scorecard

Architectural role

The scorecard assesses three separate dimensions:

- **X — Procedural:** scope, custody, provenance, permitted access and procedural compliance;
- **Y — ROMER:** reasoning transparency, evidence citation, confidence, inference control and reporting;

- **Z — Agentic:** whether the AI worked through the controlled custody system and whether its declared actions match the independent log.

The score is conjunctive rather than compensatory. A strong Y score must not conceal a weak Z score.

Scores should be assigned from the filed record and custody log, not from an impression of whether the answer looks good.

Current implementation

The rubric exists, but Session 9 could not receive a complete Z-axis score because no independent custody record was available.

Current status

Designed and partly applicable; not yet fully exercised.

The first real trace-versus-log score must await the repaired MCP-mediated rerun.

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14. Independent review

Architectural role

Independent review tests:

- whether the analytical conclusion is supported;
- whether the method remained within scope;
- whether uncertainty and absence were properly expressed;
- whether the claimed actions are supported by the infrastructure trace;
- whether the scorecard result is justified.

Current implementation

Operational through the deliberate separation of roles:

- Claude Code implements and performs the pilot work;
- Leonard/ChatGPT reviews the implementation, architecture, evidence package and conclusions.

Architectural significance

The review layer prevents the builder of the process from being its only assurer.

15. Current architecture status

Component	Current status	Principal conclusion
Human operator	Operational	Effective control of scope and material choices
Claude Code analytical agent	Operational	Strong analytical and artefact-producing capability
Meta-Primer	Partial	Principles established; formal constitutional artefact still to be fixed
CIPDA stage primer	Operational	Successfully governed the smoke-test method
ROMER template/finding	Operational	Produced a reviewable interpretive trace
MCP server	Partial	Built, but not yet proven as the actual evidence-access route
Evidence corpus	Operational	Local and read-only, but direct access remains possible
Independent custody log	Not operational in Session 9	Critical missing component
PowerShell/Blue Panel transcript	Operational	Useful operator-side record
Claude JSONL	Operational	Useful internal AI-side record
Evidence package	Operational but incomplete	Strong except for independent custody record
Defensibility scorecard	Partial	X and Y assessable; Z not yet evidenced
Independent review	Operational	Qualified-pass judgement completed

16. Where we are in the architecture

Dossier Space is no longer merely conceptual.

It currently has:

- a working operator/AI division;
- a governed local corpus;
- a functioning primer process;
- a functioning ROMER process;
- reproducible analytical artefacts;
- session and AI-side transcripts;
- independent review;
- a defined scorecard;
- a built MCP server.

What it has **not yet demonstrated** is the complete mediated architecture in which all evidence access passes through the MCP custody server and creates an independent, persistent, reconcilable log.

That distinction is central.

Session 9 proved that the **analytical architecture works**.

The next session must prove that the **custody and agentic architecture works**.

Until that occurs, Dossier Space can produce disciplined and reviewable AI analysis, but it cannot yet claim the full evidential defensibility standard the architecture is intended to provide.

17. Immediate milestone test for the next session

Before repeating the smoke test, the following should be confirmed:

1. The MCP server writes to a defined persistent log.
2. A preflight tool call appears in that log.
3. Claude's evidence access is restricted to the MCP toolset.
4. Direct Bash or filesystem access to the evidence corpus is disabled or treated as an explicit test failure.

5. Every logged access carries a session, task and finding identifier.
6. Claude separately declares the actions it believes it performed.
7. The two records are compared after the run.
8. The trace-versus-log gap is calculated.
9. The original Session 9 remains unchanged.
10. The rerun is filed as a linked, superseding test.

Milestone conclusion

We are broadly where the architecture says we should be at this point: the procedural and interpretive layers have become real and have survived a demanding evidential test.

The principal unresolved component is narrow but fundamental: **the MCP server must become the compulsory, independently logged route between the AI and the evidence, rather than merely an available tool beside direct filesystem access.**

That is the architectural threshold the next session needs to cross.